

# Professional Development and Professional Practice for Engineers - NUIST

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## Engineering Technology

May, 2026

# Professional Development and Professional Practice for Engineers - NUIST (A35623)

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|---------------------|-----------------------------------------|----------------------------|
| <b>Short Title:</b> | Prof Dev & Prof Prac for Eng<br>(NUIST) |                            |
| <b>Faculty</b>      | Science and Computing                   |                            |
| <b>Department</b>   | Engineering Technology                  |                            |
| <b>Cluster</b>      | Engineering Practise                    |                            |
| <b>Author</b>       | CLAIRE.FITZPATRICK                      |                            |
| <b>Credits</b>      | 5                                       | <b>Level:</b> Postgraduate |

## Description of Module / Aims

The professional development and Professional Practice for Engineers module is designed to develop student knowledge, skills and competencies in professional development and professional practice. It also acts as a conduit for further developing the student in areas of team work, group dynamics and tools for individual research.

## Programmes

|                                                                     |           |
|---------------------------------------------------------------------|-----------|
| MEng in Electronic Engineering (WD_EELEN_R)                         | 1 / 2 / M |
| MEng in Electronic Information Systems (WD_EELEN_RI)                | 2 / 3 / E |
| MEng in Electronic Information Systems (WD_EELEN_RI)                | 2 / 4 / E |
| MEng in Electronic Information Systems (WD_EELEN_RI)                | 3 / 5 / E |
| MEng in Electronic Information Systems (WD_EELEN_RI)                | 3 / 6 / E |
| MSc in Innovative Technology Engineering (WD_CINNT_R)               | 1 / 2 / M |
| PG Dip in Engineering in Electronic Engineering (WD_EELEN_G)        | 1 / 2 / M |
| PG Dip in Science in Innovative Technology Engineering (WD_CINNT_G) | 1 / 2 / M |

## Indicative Content

- The Professional Engineer
- Guest lectures on selected topics common with module themes from the programme
- National, European & International Professional Bodies & Regulatory Agencies
- Career Management: Self Assessment and career planning / development
- People Management Skills: Effective leadership incorporating confrontation and change management
- Technical Journal Article based on choice of project

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Identify the key characteristics of effective engineering professionals including the expected attributes of a chartered engineer.
2. Critically assess their own overall skill set and appraise their individual key strengths and weaknesses targeted to their specific professional practice.
3. Devise and implement a personal development plan comprising clear strategies for enhancing their identified existing strengths and addressing identified weaknesses.
4. Critically analyse current professional practice in relevant engineering companies.
5. Reflect on the implementation of their personal development plan and propose adjustments for future implementations.

## Learning and Teaching Methods

- Lectures
- Guest Lectures on selected topics common with module themes from the programme
- Guide to effective technical writing and academic journal submissions
- Communications workshop
- Reflective Logbook of activities
- Presentation skills

## Assessment Methods

|                       | Weighing | Outcomes Assessed |
|-----------------------|----------|-------------------|
| Continuous Assessment | 100%     |                   |
| <i>Presentation</i>   | 50%      | 1,2,3             |
| <i>Project</i>        | 25%      | 4                 |
| <i>Essay</i>          | 25%      | 5                 |

## Assessment Criteria

**70% -**

**100%:** The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

**60% - 69%:** The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

**40% - 59%:** The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

**<40%:** The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        |           |
| Practical            | 12        |           |
| Independent Learning | 99        |           |

## Supplementary Material

- .., X. "This module is based on guest lectures, discussion forums, academic papers from peer reviewed journals, on-line patent library searches and case studies." \_.\_. (.): ..